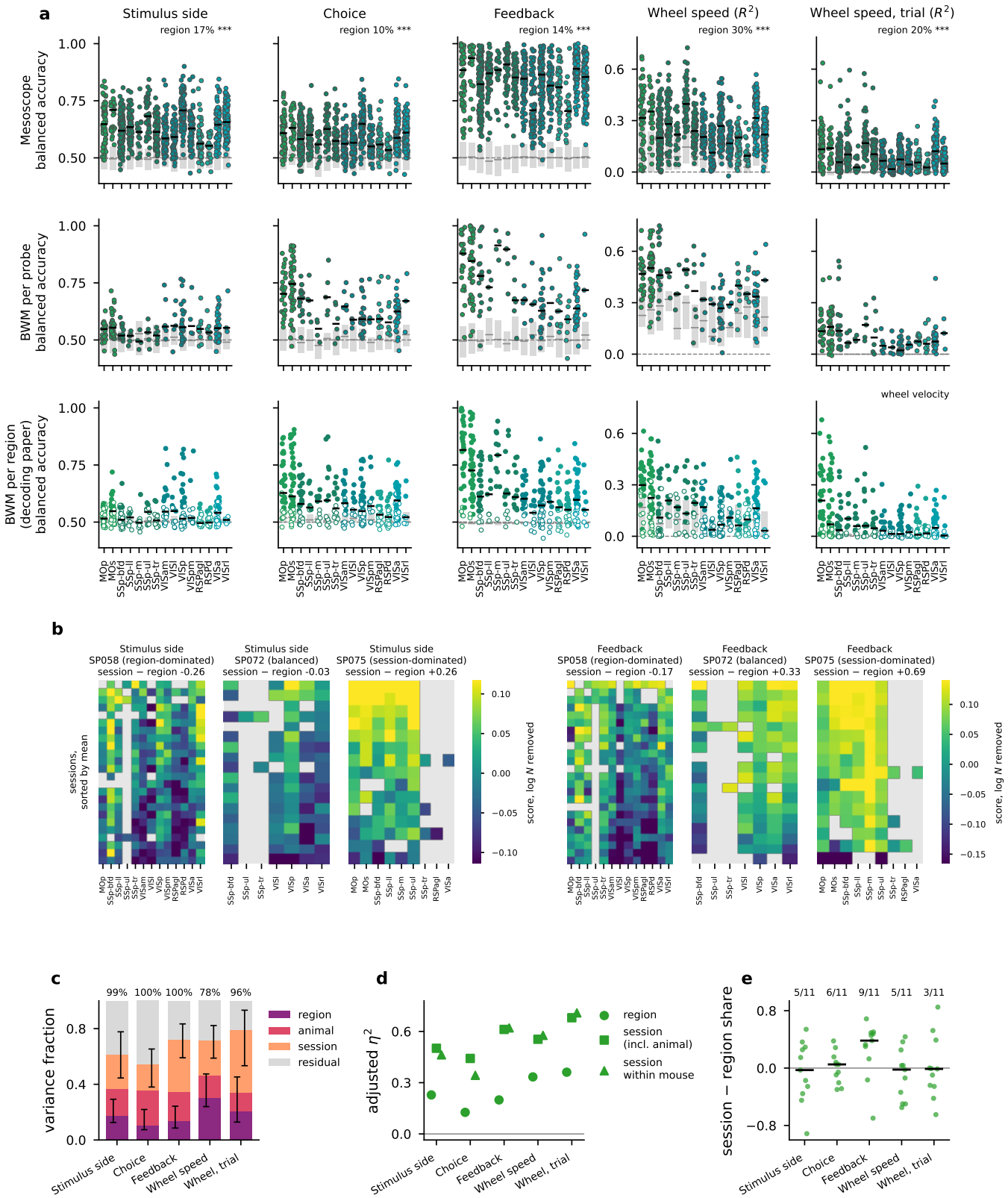


Figure 1: Variance of decoding across regions and sessions (all neurons). The 15 regions with ≥3 mesoscope sessions and BWM probes; models, tests and baselines in Methods, *Region and session variance*. **a** Scores per region (Allen order and colours; bar: median; dashed: chance; grey band and line: 5–95th percentile and median of the baseline). Rows: mesoscope, one dot per session and region (149 sessions, 12 mice); BWM, one dot per probe at its entry region (198 probes); BWM decoding paper, one dot per session and region, filled if $p < 0.05$ (wheel velocity in the last column). Inset: region share from **c**. **b** Session × region scores (log N removed) of a region-dominated, a balanced and a session-dominated mouse; grey: region not imaged. Horizontal stripes are session effects, vertical stripes region effects. **c** Mesoscope variance shares from the mixed model; error bars: 95% intervals over resampled mice; numbers: fraction of resamples with session plus animal above region. **d** Permutation tests, adjusted η^2 (0 = chance): region (circles), session including (squares) and excluding (triangles) differences between mice. Filled: $p < 0.05$. **e** Session minus region share, fitted per mouse (11 mice with ≥4 sessions); bar: median; numbers: mice with a larger session share.